

PROGRAMMING IN HASKELL

0



Topic 8.3
zipWith function

zipWith

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zipWith takes a function and two lists as parameters and then joins the two lists by applying the function between corresponding elements.

Here's how we'll implement it:

```
zipWith :: (a -> b -> c) -> [a] -> [b] -> [c]
zipWith _ [] _ = []
zipWith _ _ [] = []
zipWith f (x:xs) (y:ys) = f x y : zipWith f xs ys
```

zipWith

2

```
ghci> zipWith (+) [4,2,5,6] [2,6,2,3]  
= [6,8,7,9]
```

```
ghci> zipWith max [6,3,2,1] [7,3,1,5]  
= [7,3,2,5]
```

```
ghci> zipWith (++) ["foo ", "bar ", "buzz "]  
                ["fighters", "hoppers", "aldrin"]  
= ["foo fighters", "bar hoppers", "buzz aldrin"]
```


zipWith

3

```
ghci> zipWith (*) (replicate 5 2) [1..]  
[2,4,6,8,10]
```

```
ghci> zipWith (zipWith (*))  
      [[1,2,3],[3,5,6],[2,3,4]]  
      [[3,2,2],[3,4,5],[5,4,3]]  
  
      [[3,4,6],[9,20,30],[10,12,12]]
```

zipWith

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Note that you can use function application as the function

```
zipWith ($) funcList valueList
```

```
zipWith ($) [(+ 5),(* 3)] [1,5]  
= [6,15]
```


Aside on zipWith

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Note that you can use function application as the function

```
zipWith ($) funcList valueList
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```
zipWith ($) [(+ 5),(* 3)] [1,5]  
= [6,15]
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